

Agniveer Vayu Other than Science (Group Y) - 13 Oct 2023 - Shift-2 - Memory Based Paper

English

Question 1

Select the word from the following that is most similar in meaning to the given word.

Fascinating

Options:

- A. Clever
- B. Uniform
- C. Charming
- D. Boring

Answer: C

Solution:

The correct answer is "**Charming**"



Key Points

- Let's understand the meaning of the given words.
- **Fascinating**: It means to attract or captivate one's attention or interest. Something that is fascinating is interesting, intriguing, and holds your attention (आकर्षक).
 - For example, "The documentary on wildlife was absolutely fascinating."
- **Charming**: It means pleasing or delightful. Something that is charming has an attractive or enchanting quality (आकर्षक).
 - For example, "The old cottage in the woods was very charming."
- Thus, as per the above explanation, the most similar in meaning of the given word is "**Charming**"

Correct answer: Charming

Additional Information

- Let's understand the meaning of the other given words.
 - **Clever:** It means quick-witted, intelligent, and resourceful. It can also mean skillful or adept (चतुर).
 - For example, "My sister is very clever when it comes to solving puzzles."
 - **Uniform:** It means consistent or unvarying in form, quality, or character. It can also refer to clothing that is worn by members of an organization or group to identify themselves (एकसमान).
 - For example, "The company requires its employees to wear a uniform to work."
 - **Boring:** It means uninteresting or tedious. Something that is boring is dull, unexciting, and fails to hold your attention (उबाऊ).
 - For example, "The lecture on economics was so boring that I fell asleep in the middle."
-

Question 2

Which of the following words is an antonym for the given word?

Zealous

Options:

- A. Apathetic
- B. Enthusiastic
- C. Passionate
- D. Eager

Answer: A

Solution:

The correct answer is "Apathetic".

Key Points

- 'Zealous' means full of, characterized by, or due to zeal; ardently active, devoted, or diligent. It implies showing intense enthusiasm or passion for something. ('उत्साही')
 - **Example-** "The scientist was zealous in his pursuit of new discoveries."
- 'Apathetic' means showing or feeling no interest, enthusiasm, or concern ('उदासीन')
 - **Example-** "He was apathetic towards the sufferings of others."
- Hence, the option that is opposite in meaning to the given word is 'Apathetic'.
- Therefore, the correct answer is 'Option 1'.

Additional Information

- Let's have a look at the meaning of the remaining options.
 - '**Enthusiastic**' means showing intense and eager enjoyment, interest, or approval.
 - '**Passionate**' refers to having, showing, or caused by strong feelings or beliefs.
 - '**Eager**' is wanting to do or have something very much.
-

Question 3

Change the given sentence in Indirect speech:

Teacher said, "Oil floats on water."

Options:

- A. The teacher said that oil floats on water.
- B. The teacher said that oil floating on water.
- C. The teacher told oil floats on water.
- D. The teacher spoke that oil floats on water.

Answer: A

Solution:

The correct answer is "**The teacher said that oil floats on water.**"

Key Points

- The structure of reported speech involves changing the original speaker's words and possibly the verb tense, depending on the context.
 - However, in this case, since the statement made by the teacher is **a fact or general truth** ("**Oil floats on water**"), the tense remains unchanged.
- In the direct speech sentence, "**Teacher said, 'Oil floats on water'**", the teacher is the subject and '**said**' is the reporting verb.
- The words spoken by the teacher, also known as the reported clause, are "**Oil floats on water.**"
- When converting this into indirect (or reported) speech, we introduce a conjunction ('**that**', in this case) to join the reporting clause with the reported clause.
- The sentence then becomes: "**The teacher said that oil floats on water.**"
- Here's the breakdown:
 - **Direct Speech:** Teacher said, "Oil floats on water."
 - **Indirect Speech:** The teacher said that oil floats on water.
- This indicates that the teacher stated the fact that oil floats on water.

Hence, The correct answer is "**Option 1**".

Question 4

Change the given sentence in passive voice:

I did not speak even a single word in the meeting.

Options:

- A. Not even a single word in the meeting was spoken by me.
- B. In the meeting did not speak even a single word by me.
- C. Even a single word was not spoken by me in the meeting.
- D. A meeting was not spoken by a single word of me.

Answer: C

Solution:

The correct answer is "**Even a single word was not spoken by me in the meeting.**"



Key Points

- The given sentence is in active voice.
- To change it to passive voice, the object of the verb becomes the subject, and the verb is changed to its past participle form.
- The auxiliary verb '**was/were**' is used according to the number of the subject.
- In this case, the object is '**even a single word**', which becomes the subject in the passive voice.
- The verb '**speak**' is changed to its past participle form '**spoken**'.
- Therefore, the correct answer is '**Even a single word was not spoken by me in the meeting.**'

Hence, the correct answer is "**Option 3**".



Additional Information

- Option (1) is incorrect because it uses the active voice construction 'was spoken by me'.
 - Option (2) is incorrect because it is grammatically incorrect.
 - Option (4) is incorrect because it is grammatically incorrect and does not make sense.
-

Question 5

Select the incorrectly spelt word.

Options:

- A. Beuatiful
- B. Irreplaceable
- C. Appearance
- D. Grammar

Answer: A

Solution:

The correct answer is **option 1 i.e. beuatiful**.



Key Points

- Out of all the options, only '**beuatiful**' is incorrectly spelled.
- **Beautiful** is the correct spelling of 'beuatiful'. Beautiful means pleasing the senses or mind aesthetically.
- - Eg- *Neha is a beautiful girl.*



Additional Information

- Correction of other words given in the option and their meanings are as follows:
 - **Appearance** is the correct spelling that means the way that someone or something looks.
 - **Grammar** is the correct spelling which is the way we arrange words to make proper sentences.
-

Question 6

Select the **INCORRECTLY** spelt word.

Options:

- A. Alien
- B. Awfull

C. Across

D. Already

Answer: B

Solution:

The correct answer is **Option (2)** i.e. 'Awfull'.



Key Points

- Awful is the correct spelling of 'awfull'.
- Awful : (भयानक) - Very bad or unpleasant.
 - **Example:** The weather today is awful.

Therefore, the correct answer is 'Awful'.



Additional Information

- Let's explore the other options:
 - Alien : (अलग, विदेशी) - Unfamiliar and disturbing or distasteful.
 - **Example:** The custom was alien to him.
 - Across : (पार) - From one side to the other.
 - **Example:** He walked across the road.
 - Already : (पहले से) - Prior to a specified or implied time.
 - **Example:** She is already here.

Question 7

What is the noun form of 'successful'?

Options:

A. succeed

B. successor

C. success

D. Successfulness

Answer: C

Solution:

The correct answer is "**Success**".

Key Points

- 'Successful' is an adjective that is used to describe something or someone that has accomplished desired aims or attained prosperity. For example, a successful business, a successful negotiation, etc. So, it's something we use to describe nouns (people, places, things, ideas, etc.).
- 'Success', on the other hand, is a noun. As the desired noun form of 'successful', it describes the accomplishment of an aim or purpose. For instance, an achievement of an objective is a 'success'. The word is often used in various contexts to indicate the state or condition of meeting a defined range of expectations. So in the general context, it indicates a notable accomplishment.
- The transformation of words between their various forms, in this case from an adjective to a noun, is a common practice in English and it's simply a matter of knowing how each form of the word is used. This kind of transformation helps us expand our vocabulary and provides us with a more sophisticated and nuanced way to express ourselves.

Hence, the correct answer is '**Success**'.

Question 8

In the following sentence, only one part has an error. Choose the part with the error.

Many a man donot know his own good deeds.

Options:

- A. Many a man
- B. do not know
- C. his
- D. own good deeds.

Answer: B

Solution:

The correct answer is '**do not know**'.

Key Points

- The error lies in the usage of '**do not know**' instead of '**does not know**' in the 2nd segment of the given sentence.

- In the given sentence, we are talking about a *large number of people who don't know about their own good deeds*.
- **Many a (formal)** phrase is used with a **singular noun** and **verb** to mean 'a large number of'.
- For example: - **Many a good man has** been destroyed by drink.
- Hence, the correct answer is **option 2**.

Correct Sentence: *Many a man does not know his own good deeds.*

Question 9

Following sentences have an error in one of its parts. Find the part which has an error and mark it accordingly.

All those (A) /who feel that (B) /the country has progressed (C)/ contributes to it progress (D)

Options:

A. D

B. C

C. A

D. B

Answer: A

Solution:

The correct answer is **Option 1**.

Key Points

- The primary issue in the sentence is the incorrect subject-verb agreement in **part D**.
- The subject of this part refers to "**all those who feel**," which is plural, so the verb should also be plural ("**contribute**" instead of "**contributes**").
- Additionally, "**it**" should be the possessive pronoun "**its**" to correctly indicate possession.
- Thus, the error is located in **part D** of the sentence, which should be corrected to "contribute to its progress."

Therefore, the correct answer is **D**.

Correct Sentence: *All those who feel that the country has progressed **contribute to its progress**.*

Additional Information

- **A: "All those"** – This part of the sentence is grammatically correct and does not require any changes.
 - **B: "who feel that"** – This part of the sentence is grammatically correct and does not require any changes.
 - **C: "the country has progressed"** – This part of the sentence is grammatically correct and does not require any changes.
-

Question 10

Improve the bracketed part in the given sentence.

The judge (will heard) his case tomorrow.

Options:

- A. will hear
- B. will hearing
- C. will hears
- D. will hearing

Answer: A

Solution:

The correct answer is "**will hear**".

Key Points

- The problem with the original sentence, "The judge will heard his case tomorrow", is that it misuses the future tense. In English, the future tense of a verb is generally formed by using the auxiliary verb "will" together with the base form of a main verb.
- "Will heard" is incorrect because "heard" is the past tense of the verb "hear". The future tense should be formed with "will" followed by the base form of the verb which is "hear". So the verb that should be used here is "**hear**".
- Option B and D, "will hearing" and "will hearing" don't make sense as there is no such word as "hearing" and "hearing" is a gerund or present participle of the verb "hear", not appropriate after "will" in a future tense structure.
- Option C "will hears" is incorrect because the base form of the verb "hear" is used after "will", not its third person present form "hears".

Hence, the correct answer is "**Option 1**".

Question 11

What is the most appropriate meaning of the idiom?

Burn the midnight oil

Options:

- A. To set something on fire
- B. To stay up and work late into the night
- C. To waste resources
- D. To confront a fear

Answer: B

Solution:

The correct answer is “**To stay up and work late into the night**”.



Key Points

- **Burn the midnight oil** means to stay up and work late into the night.
- The phrase metaphorically refers to hard work and dedication.
- It highlights a person's willingness to sacrifice sleep to get work done or meet deadlines.
 - **Example-** I had to **burn the midnight oil** to complete the project on time."

Hence, the correct answer is ‘**Option 2**’.

Question 12

Choose the word that can substitute the given sentence.

Soldiers who fight on horse back

Options:

- A. Captain

B. Pilot

C. Major

D. Cavalry

Answer: D

Solution:

The correct answer is '**Cavalry**'.



Key Points

- **Cavalry** means the group of soldiers in an army who fight in tanks, or (especially in the past) on horses. (घुड़सवार सेना)
 - **Example:** The **cavalry** charged up the hill.
- Some **synonyms** of the word '**cavalry**' are '**Troopers**', '**lancers**', '**dragoons**'.
- Therefore, the most appropriate answer of the given meaning is '**cavalry**'.
- Hence, **option 4** is the correct answer.



Additional Information

Let's learn about the other given options:

- A **captain** is a person in command of a ship. (एक जहाज की कमान में व्यक्ति।)
- A **pilot** is a person who operates the flying controls of an aircraft. (एक व्यक्ति जो एक विमान के उड़ान नियंत्रण को संचालित करता है।)
- **Major:** is a commissioned military officer ranking next below a lieutenant colonel and next above a captain lieutenant colonel. (सैन्य अधिकारी की रैंकिंग लेफ्टिनेंट कर्नल से नीचे और कैप्टन लेफ्टिनेंट कर्नल से ऊपर होती है।)

Question 13

Fill in the blanks with the correct option:

If I had told the truth, _____ the punishment.

Options:

A. I would have escaped

B. I will escape

C. I would escape

D. I will not escape

Answer: A

Solution:

The correct answer is '**I would have escaped**'.



Key Points

- The given sentence is an example of **Type three conditional sentences**.
- This type of conditional sentence indicates that "if" something would have happened, something else could have followed.
- The "if" clause in third conditionals is in the past perfect verb tense because it has already happened.
- The result clause still uses "would," but adds "have" for the perfect conditional verb tense "would have + verb."
 - Example -*If Bonnie had studied, she would have passed the test.*
- In the given sentence, it is evident that if he had told the truth, he could have escaped punishment.
- Thus, '**I would have escaped**' is the correct option to fill in the blank.

Hence, the correct answer is **option (1)**.



Additional Information

- The conditional mood is one of five sentence moods in grammar. It uses the conjunction "if" (and occasionally "when") to express a condition and its result.
- **Zero Conditional Sentences** -The outcome in a zero conditional sentence is certain. It is used when one thing always causes another.
 - Example -*If you trip on pavement, you get a scrape.*
- **First Conditional Sentences** -When a situation is not completely certain, it's time for first conditionals. First conditionals (or Type 1 conditionals) dictate that a result is not certain, but very likely that one condition will lead to another.
 - Example -*I will answer if he calls me.*
- **Second Conditional Sentences** -Second conditionals use the modal verb "would" to indicate that it is unlikely that a condition will be fulfilled.
 - Example -*I would answer if he called.*

Question 14

Choose the correct conjunction to fill in the blank:

I was hungry, ____ I ate the apple.

Options:

A. and

B. yet

C. so

D. or

Answer: C

Solution:

The correct answer is "So".



Key Points

- The conjunction "so" is often used to indicate a result or an effect.
- In this sentence, it shows the cause-effect relationship between the speaker's state of hunger and the action of eating the apple.
- The person eats the apple because he or she is hungry.
- Therefore, "so" is the best option as it properly indicates this cause-effect relationship.
- The correct sentence is-**I was hungry, so I ate the apple.**

Hence, the correct answer is 'Option 3'.

Question 15

Fill in the blank with the most suitable word.

She ran because she _____ in a hurry.

Options:

A. is

B. was

C. must be

D. could be

Answer: B

Solution:

The correct answer is 'was'.

Key Points

- Given the structure and context of the sentence, we are dealing with past tense narrative. "She ran" is in the past tense.
- The sentence is about a past event where the subject was in a hurry, which makes 'was' the appropriate tense.
- 'She' is a singular subject, so the verb following 'she' should also be in its singular form in the past tense.
- 'Was' correctly associates the act of running with the state of the character being in a hurry.
- The presence of the word 'because' derives that the mood is indicative and something factual is being communicated. Thus, the verb that fits here should keep the mood and tone factual, which 'was' does accurately.

Therefore, the correct answer is Option 2.

Additional Information

- 'is' would be used if the action is in the present, which is not the case here.
- 'must be' implies an ongoing assumption or necessity rather than a factual state.
- 'could be' suggests a possibility or potential situation, not a fact.

Question 16

Direction: Read the passage carefully and answer the question that follows.

Advances in medical science and technology have so thoroughly overtaken our lives at the turn of the millennium that it is difficult to imagine what else we might see and encounter in our lifetimes. Incredible journeys have now gone beyond mere geography; scientifically, we have journeyed for inside our own bodies. Biotechnology has transformed from boring laboratory science to a lively and pioneering branch of science. Some of the recently developed biotechnologies, such as in vitro fertilisation and genetic modification, xenotransplantation and cloning, appear straight out of bizarre science fiction and promise unthinkable potential possibilities. While understanding more about the building blocks of plants will declare a food revolution on a scale larger than the Green Revolution of the 1960s and 1970s, learning more about the human genome would open up newer and more detailed techniques for overcoming disease and debilitation.

The novel A Space Odyssey, based on Arthur Clarke's novel of the same name, alluded to the celestial essence of human evolution. On the one hand, we've discovered 6-billion-year-old bacteria that are still alive, which scientists believe may have been thrown into the planet by debris from a falling meteorite from space. Space probes, on the other hand, have discovered evidence that Mars once had water. With the establishment of an international space station and more regular space flights, we could be able to successfully communicate life forms in other parts of the universe.

We have progressed from micro and laser surgery to surgery facilitated by precision-working robot assistants.

What are the advantages of learning more about the human genome?

Options:

- A. It will lead to the creation of newer, more detailed methods for treating illness and debilitation.
- B. It will facilitate inmicro and laser surgeries.
- C. It can aid in the understanding of in vitro fertilisation and genetic modification.
- D. All the above

Answer: A

Solution:

The correct answer is '**It will lead to the creation of newer, more detailed methods for treating illness and debilitation.**'



Key Points

- Let's refer to the passage, "learning more about the **human genome** would open up newer and more detailed techniques for **overcoming disease and debilitation.**"
 - From the given lines, it can be understood that learning more about the **human genome** will open up more **in-depth solutions to overcome illness and weakness.**
 - Therefore, **option 1** is the correct answer.
-

Question 17

Direction: Read the passage carefully and answer the question that follows.

Advances in medical science and technology have so thoroughly overtaken our lives at the turn of the millennium that it is difficult to imagine what else we might see and encounter in our lifetimes.

Incredible journeys have now gone beyond mere geography; scientifically, we have journeyed for inside our own bodies. Biotechnology has transformed from boring laboratory science to a lively and pioneering branch of science. Some of the recently developed biotechnologies, such as in vitro fertilisation and genetic modification, xenotransplantation and cloning, appear straight out of bizarre science fiction and promise unthinkable potential possibilities. While understanding more about the building blocks of plants will declare a food revolution on a scale larger than the Green Revolution of the 1960s and 1970s, learning more about the human genome would open up newer and more detailed techniques for overcoming disease and debilitation.

The novel A Space Odyssey, based on Arthur Clarke's novel of the same name, alluded to the celestial essence of human evolution. On the one hand, we've discovered 6-billion-year-old bacteria that are still alive, which scientists believe may have been thrown into the planet by debris from a falling meteorite from space. Space probes, on the other hand, have discovered evidence that Mars once had water. With the establishment of an international space station and more regular space flights, we could be able to successfully communicate life forms in other parts of the universe.

We have progressed from micro and laser surgery to surgery facilitated by precision-working robot assistants.

What do scientists think of bacteria that are 6 billion years old?

Options:

- A. These bacteria, according to scientists, are no longer alive.
- B. Scientists believe that the bacteria may have been thrown into the planet by debris from a falling meteorite from space.
- C. Scientists assume that the bacteria was discovered by modern humans.
- D. Scientists believe that the bacteria may have existed even before the human evolution.

Answer: B

Solution:

The correct answer is 'Scientists believe that the bacteria may have been thrown into the planet by debris from a falling meteorite from space.'



Key Points

- Let's refer to the passage, "On the one hand, we've discovered **6-billion-year-old bacteria that are still alive, which scientists believe may have been thrown into the planet by debris from a falling meteorite from space.**"
 - From the given lines, it can be understood that Scientists assume that **6-billion-year-old bacteria** that are still alive were **thrown into the atmosphere by debris** from a falling meteorite from space.
 - Therefore, **option 2** is the correct answer.
-

Question 18

Direction: Read the passage carefully and answer the question that follows.

Advances in medical science and technology have so thoroughly overtaken our lives at the turn of the millennium that it is difficult to imagine what else we might see and encounter in our lifetimes. Incredible journeys have now gone beyond mere geography; scientifically, we have journeyed for inside our own bodies. Biotechnology has transformed from boring laboratory science to a lively and pioneering branch of science. Some of the recently developed biotechnologies, such as in vitro fertilisation and genetic modification, xenotransplantation and cloning, appear straight out of bizarre science fiction and promise unthinkable potential possibilities. While understanding more about the building blocks of plants will declare a food revolution on a scale larger than the Green Revolution of the 1960s and 1970s, learning more about the human genome would open up newer and more detailed techniques for overcoming disease and debilitation.

The novel *A Space Odyssey*, based on Arthur Clarke's novel of the same name, alluded to the celestial essence of human evolution. On the one hand, we've discovered 6-billion-year-old bacteria that are still alive, which scientists believe may have been thrown into the planet by debris from a falling meteorite from space. Space probes, on the other hand, have discovered evidence that Mars once had water. With the establishment of an international space station and more regular space flights, we could be able to successfully communicate life forms in other parts of the universe.

We have progressed from micro and laser surgery to surgery facilitated by precision-working robot assistants.

The novel *A Space Odyssey* referred to which of the following?

Options:

- A. bacteria that are still alive
- B. science fiction
- C. the celestial essence of human evolution
- D. life forms in other parts of the universe.

Answer: C

Solution:

The correct answer is '**the celestial essence of human evolution**'.



Key Points

- Let's refer to the passage, "The novel *A Space Odyssey*, based on Arthur Clarke's novel of the same name, **alluded to the celestial essence of human evolution.**"
 - From the given lines, it can be understood that based on Arthur Clarke's novel, the novel "*A Space Odyssey*", referred to **the celestial essence of human evolution.**
 - Therefore, **option 3** is the correct answer.
-

Question 19

Direction: Read the passage carefully and answer the question that follows.

Advances in medical science and technology have so thoroughly overtaken our lives at the turn of the millennium that it is difficult to imagine what else we might see and encounter in our lifetimes.

Incredible journeys have now gone beyond mere geography; scientifically, we have journeyed for inside our own bodies. Biotechnology has transformed from boring laboratory science to a lively and pioneering branch of science. Some of the recently developed biotechnologies, such as in vitro fertilisation and genetic modification, xenotransplantation and cloning, appear straight out of bizarre science fiction and promise unthinkable potential possibilities. While understanding more about the building blocks of plants will declare a food revolution on a

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We have progressed from micro and laser surgery to surgery facilitated by precision-working robot assistants.

What did the space probes discover?

Options:

- A. evidence that Mars has life in it
- B. evidence that Mars has water
- C. evidence that Mars has 6-billion-year-old bacteria that are still alive
- D. evidence that Mars once had water

Answer: D

Solution:

The correct answer is 'evidence that Mars once had water'.

Key Points

- Let's refer to the passage, "**Space probes**, on the other hand, have discovered **evidence that Mars once had water.**"
- Therefore, from the given lines, it can be concluded that **option 4** is the correct answer.

Additional Information

- **Meanings of difficult words from the given paragraph:**
 - **alluded:** suggest or call attention to indirectly; hint at.
 - **debris:** loose natural material consisting especially of broken pieces of rock.
 - **precision:** the quality, condition, or fact of being exact and accurate.

Question 20

Direction: Read the passage carefully and answer the question that follows.

Advances in medical science and technology have so thoroughly overtaken our lives at the turn of the millennium that it is difficult to imagine what else we might see and encounter in our lifetimes.

Incredible journeys have now gone beyond mere geography; scientifically, we have journeyed for inside our own bodies. Biotechnology has transformed from boring laboratory science to a lively and pioneering branch of science. Some of the recently developed biotechnologies, such as in vitro fertilisation and genetic modification, xenotransplantation and cloning, appear straight out of bizarre science fiction and promise unthinkable potential possibilities. While understanding more about the building blocks of plants will declare a food revolution on a scale larger than the Green Revolution of the 1960s and 1970s, learning more about the human genome would open up newer and more detailed techniques for overcoming disease and debilitation.

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We have progressed from micro and laser surgery to surgery facilitated by precision-working robot assistants.

Which of the following is/are true considering Biotechnology?

Options:

- A. Biotechnology has evolved from a gloomy laboratory science to a vibrant or pioneering branch of science.
- B. In vitro fertilisation is a relatively new biotechnology.
- C. Genetic modification, xenotransplantation and cloning are some of the recently developed biotechnologies.
- D. All the above

Answer: D

Solution:

The correct answer is 'All the above'.

Key Points

- Let's refer to the given passage, "**Biotechnology** has transformed from **boring laboratory** science to a **lively and pioneering branch of science**. Some of the **recently developed biotechnologies**, such as **in vitro fertilisation and genetic modification, xenotransplantation and cloning**, appear straight out of bizarre science fiction and promise unthinkable potential possibilities."
- From the given lines, it can be understood that Biotechnology **has progressed to a pioneering branch of science** and some of the most recently established biotechnologies include **in vitro fertilisation, genetic engineering, xenotransplantation, and cloning**.
- Therefore, **option 4 'all the above'** is the correct answer.

RAGA

Question 21

Kumar's father Arun's son Shankar's wife Hema is the daughter in law of Kumar's father's wife Prema. How is Prema related to Kumar?

Options:

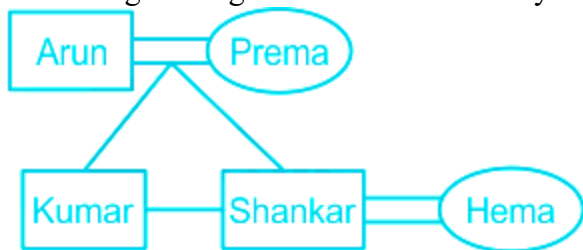
- A. Mother
- B. Daughter
- C. Father
- D. Son

Answer: A

Solution:

Symbol in Diagram	Meaning
○	Female
□	Male
==	Married Couple
—	Siblings
	Difference of A Generation

According to the given information family tree will be,



Prema is **mother** of Kumar.

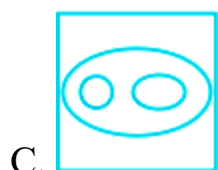
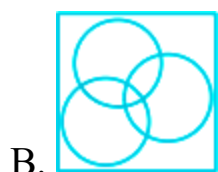
Hence, "**Option 1**" is the correct answer.

Question 22

Select the Venn diagram that best represents the relationship between the following classes.

Animals, Cows, Dogs

Options:



Answer: C

Solution:

According to the given information:

Cow and dogs are animal.

But they are different from each other.

So, the Venn diagram for them will be the circle as a animal class consisting two circle, one of dog and other of cat.

Hence, Option 3 is correct.

Question 23

For a standard clock, what is the angle between hour hand and the minute hand when the clock shows 4:30 am?

Options:

A. 45°

B. 45.5°

C. 27.6°

D. 40°

Answer: A

Solution:

The angle between hour hand and the minute hand when the clock shows 4:30 am can be calculated by using the following formula:

$$\text{Angle} = \frac{11}{2} \times M - 30 \times H$$

Where, M = Minutes and H = Hours

Here M = 30 and H = 4

Therefore, we get angle = $\frac{11}{2} \times 30 - 30 \times 4 = 165 - 120 = 45^\circ$.

Hence, **45°** is the correct answer.

Question 24

Select the figure from among the given options that can replace the question mark (?) in the following series.



Options:

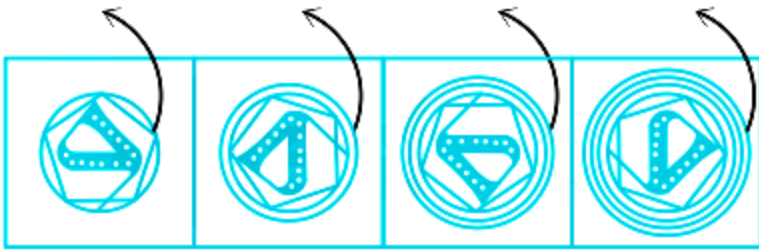


Answer: A

Solution:

The pattern followed here is;

Inner figure is Rotated 45° anti clockwise and one circle is increased in each step.



So, the next image is;



Hence, the correct answer is "**Option (1)**".

Question 25

select the figure from the given options that can replace the question mark (?) in the following series.



Options:



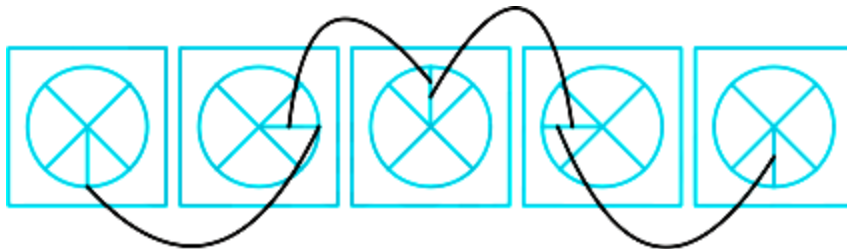


Answer: A

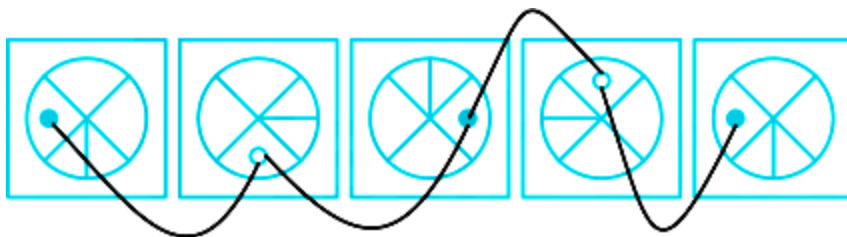
Solution:

The logic followed here is:

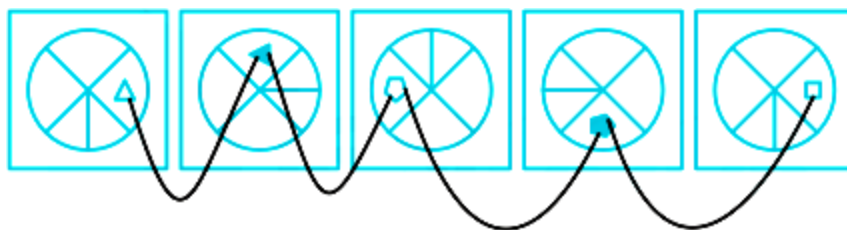
Step I: A Straight line given in a part of circle has shifted to its next place in next image.



Step II: Circle are shifted to its next place and shaded in alternate image.



Step III: The Polygons are shaded in each alternate image.



So, 'Option 1' will replace the question mark.

Hence, the correct answer is **"Option 1"**.

Question 26

Choose one of the possibilities to represent the next figure in the series.



Options:

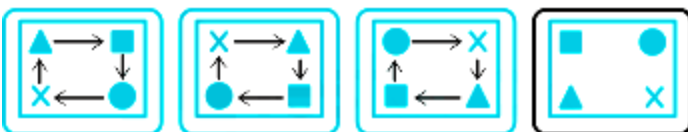


Answer: C

Solution:

The figure that will replace the question mark (?) in the following figure series is shown below:

All shapes rotate in the clockwise direction according to the given arrow in the figure given below.



Hence, 'option 3' is the correct answer.

Question 27

Choose one of the possibilities to represent the next figure in the series.



Options:

A.



B.



C.



D.



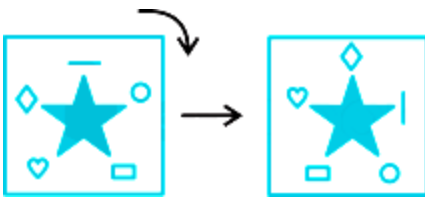
Answer: C

Solution:

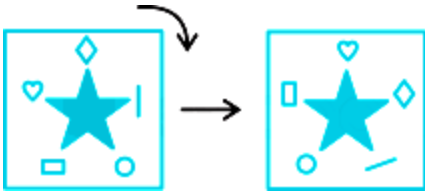
The logic followed in the given figure series:

Logic: 'Rotation of figure in clockwise direction'.

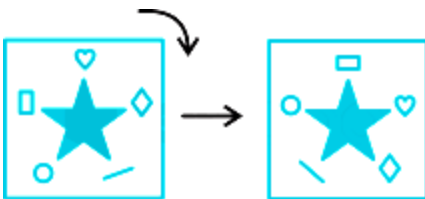
From Figure (1) to (2):



From Figure (2) to (3):



From Figure (3) to answer figure:



Hence, the correct answer is "Option (3)".

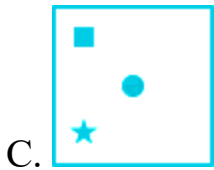
Question 28

Select the figure that will replace the question mark (?) in the following figure series.



Options:



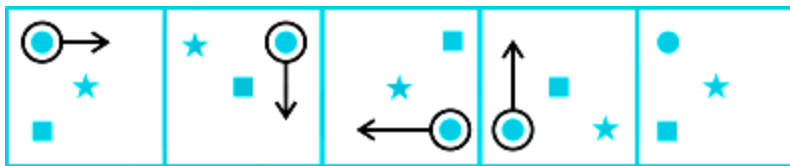


Answer: B

Solution:

The figure that will replace the question mark (?) in the following figure series is shown below:

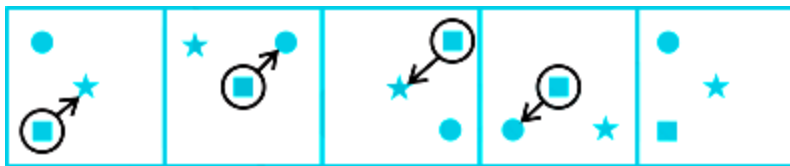
1) For '●' element : Shifting in the clockwise directions subsequently.



2) For '★' element : Shifting diagonally to left corner upwards then right corner downwards directions subsequently.



3) For '■' element : Shifting diagonally to right corner upwards then left corner downwards directions subsequently.



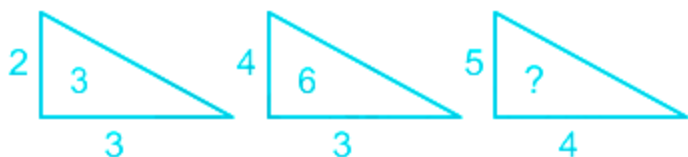
Final series is :



Hence, 'option - (2)' is the correct answer.

Question 29

Assuming that the characters in the set follows a certain pattern, find the missing character.



Options:

A. 12

B. 8

C. 16

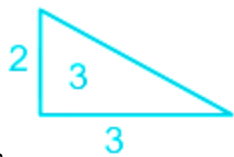
D. 10

Answer: D

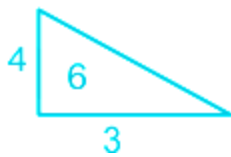
Solution:

The logic followed here is:

Logic: Product of the outer numbers $\div 2$ = Inner number in the triangle.

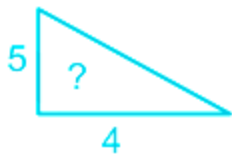


For, $\rightarrow (2 \times 3) \div 2 \rightarrow 6 \div 2 = 3.$



For, $\rightarrow (4 \times 3) \div 2 \rightarrow 12 \div 2 = 6.$

Similarly,



$\rightarrow (5 \times 4) \div 2 \rightarrow 20 \div 2 = 10.$

Hence, the correct answer is "Option 4".

Question 30

Select the correct option that indicates the arrangement of following units in a logical and meaningful order.

1. Byte

2. Megabyte

3. Terabyte

4. Gigabyte

5. Kilobyte

Options:

A. 3, 1, 2, 4, 5

B. 1, 5, 4, 2, 3

C. 1, 5, 2, 4, 3

D. 1, 5, 2, 3, 4

Answer: C

Solution:

Logic: Arranged in ascending order.

Byte < Kilobytes (KB) < Megabyte (MB) < Gigabyte (GB) < Terabyte (TB)

Hence, the correct answer is "option (3)".



Additional Information

Byte - 8 bits

Kilobyte (KB)- 1024bytes

Megabyte (MB) - 1024Kilobytes (KB)

Gigabyte (GB) - 1024Megabytes (MB)

Terabyte (TB) - 1024Gigabytes (GB)

Question 31

P, Q, R, S, T, U and V are seven friends having different heights. R is taller than 4 persons. Q is taller than R but shorter than P. T is taller than U but shorter than S. U is not the shortest. How many people are shorter than T?

Options:

A. 3

B. 5

C. 2

D. 1

Answer: C

Solution:

Given:P, Q, R, S, T, U and V are seven friends having different heights.

1) R is taller than 4 persons.

___ > ___ > R > ___ > ___ > ___

2) Q is taller than R but shorter than P.

P > Q > R > ___ > ___ > ___

3) T is taller than U but shorter than S.

4) U is not the shortest.

P > Q > R > S > T > U > V

So, there are 'two' people are shorter than T.

Hence, the correct answer is **"Option 3"**.

Question 32

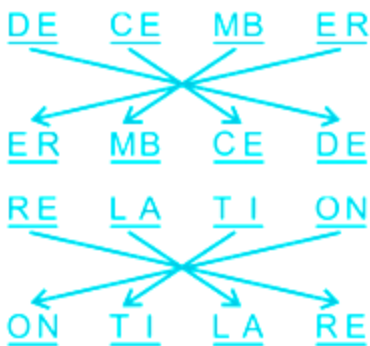
In a certain language, DECEMBER is written as ERMBCED E. Find the code for RELATION.

Options:

- A. ONTILARE
- B. SERWGSNI
- C. OMTJAREL
- D. PRWHHSNI

Answer: A

Solution:



Hence, ONTILARE is the correct answer.

Question 33

Six friends are sitting in a circle facing the centre. D is in between P and Q. T is between M and L, P and M are exactly opposite to each other. P is to the immediate right of L. Who is sitting exactly opposite to Q?

Options:

- A. P

B. L

C. D

D. M

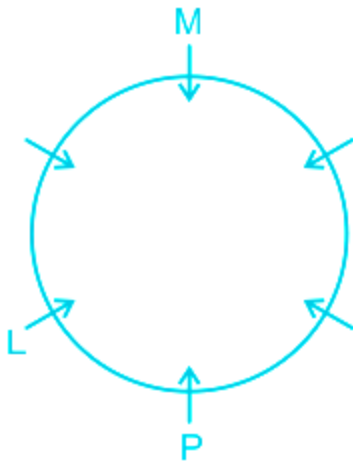
Answer: B

Solution:

Given: Six friends are sitting in a circle facing the centre.

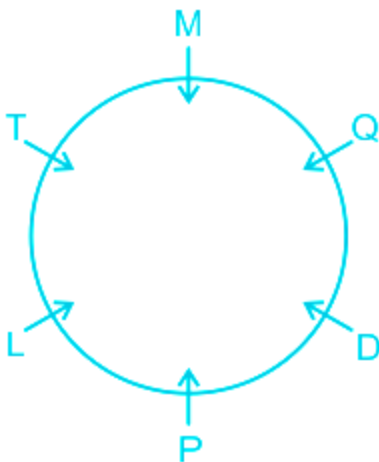
1) P and M are exactly opposite to each other.

2) P is to the immediate right of L.



3) T is between M and L.

4) D is in between P and Q.



Here, 'L' is sitting exactly opposite to Q.

Hence, the correct answer is **"Option 2"**.

Question 34

Vikas started walking towards East. He took a left turn after walking 2 km. He again took a left turn after walking 5 km. Which direction is he facing now?

Options:

- A. West
- B. South
- C. North
- D. East

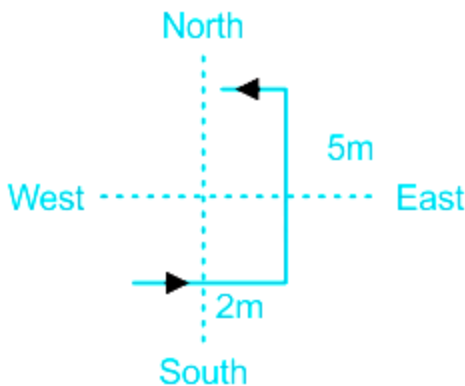
Answer: A

Solution:

Given:

- 1) Vikas started walking towards East.
- 2) He took a left turn after walking 2 km.
- 3) He again took a left turn after walking 5 km.

Distance traveled by Vikas:



Now, he is facing in the West direction.

Hence, the correct answer is West.

Question 35

Select the number from among the given options that can replace the question mark (?) in the following series.

5, 10, 8, 16, 14, 28, ?

Options:

A. 30

B. 26

C. 56

D. 24

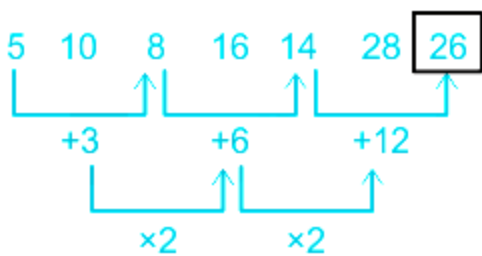
Answer: B

Solution:

Given series:

5, 10, 8, 16, 14, 28, ?

The logic followed in the given series:



Here, '26' is the next term as per the logic.

Hence, the correct answer is "Option (2)".

Question 36

The ratio between two numbers is 2 : 3. If their LCM is 150. Then the numbers are

Options:

- A. 30, 40
- B. 48, 64
- C. 50, 75
- D. None of these

Answer: C

Solution:

ConceptUsed:

LCM -Smallest multiplethat two numbers have in common.

Calculation:

Let two numbers be $2x$ & $3x$ respectively.

$$\text{LCM}(2x, 3x) = 6x$$

$$\Rightarrow 6x = 150$$

$$\Rightarrow x = 25$$

$$\Rightarrow 2x = 2 \times 25 = 50, 3x = 3 \times 25 = 75$$

$\therefore$ The numbers are 50 and 75.

Question 37

What is the value of $\sin^2 25^\circ + \sin^2 65^\circ$?

Options:

- A. 0
- B. 1
- C. 2
- D. 4

Answer: B

Solution:

Formula used:

$$\sin^2\theta + \cos^2\theta = 1$$

Calculation:

$$\sin^2 25^\circ + \sin^2 65^\circ$$

$$\Rightarrow \sin^2 25^\circ + \sin^2 (90^\circ - 25^\circ)$$

$$\Rightarrow \sin^2 25^\circ + \cos^2 25^\circ$$

$$\Rightarrow 1$$

$\therefore$ The required value is 1.

Question 38

If two-third of goods are sold at 25% profit, 20% are sold at 20% loss and rest at a profit of 20%, then there is a profit of Rs. 3312. The cost price of the goods is:

Options:

A. Rs. 21400

B. Rs. 20000

C. Rs. 21600

D. Rs. 18600

Answer: C

Solution:

Let there are 150 goods cost of each be 1 unit, then

Total CP of 150 goods = 150 unit

$\Rightarrow \frac{2}{3}$ of 150 = 100, 20% of 150 = 30 and Remaining goods = 150 - 100 - 30 = 20

Total SP of 150 goods = $100 \times \frac{5}{4} + 30 \times \frac{4}{5} + 20 \times \frac{6}{5} = 125 + 24 + 24 = 173$ unit

Profit = 173 - 150 = 23 unit

$\Rightarrow 23 \text{ unit} = 3312$

$\Rightarrow 1 \text{ unit} = 144$

$\Rightarrow 150 \text{ unit} = 150 \times 144 = \text{Rs. } 21600$

$\therefore$ Cost of price of goods is Rs. 21600.

Question 39

The perimeter of the floor of a room is 18 m. What is the area of the walls of the room, if the height is 3 m?

Options:

A. 21 m²

B. 42 m²

C. 54 m²

D. 108 m²

Answer: C

Solution:

Given:

Perimeter of the floor = 18 m

Height of room = 3m

Formula Used:

Area of the four walls of cuboid $2(l + b)h$

Where, $l \rightarrow$ Length, $b \rightarrow$ Breadth and $h \rightarrow$ Height

Calculation:

Let the length and breadth of a room be l and b .

$$\text{Perimeter of floor} = 2(l + b)$$

$$\Rightarrow 2(l + b) = 18$$

$$\Rightarrow l + b = 9 \text{ m}$$

$$\text{Area of walls} = 2(l + b)h$$

$$\Rightarrow 2 \times 9 \times 3$$

$$\Rightarrow 54 \text{ m}^2$$

$\therefore$ The area of the four walls of the room is 54 m^2 .

Question 40

Two-fifth of one-third of three-seventh of a number is 18. What is 60% of that number?

Options:

A. 190

B. 191

C. 192

D. 189

Answer: D

Solution:

Given:

Two-fifth of one-third of three-seventh of a number is 18

Calculations:

Let the unknown term be x

$$\frac{2}{5} \text{ of } \frac{1}{3} \text{ of } \frac{3}{7} \text{ of } x = 18$$

$$\Rightarrow \frac{2x}{5 \times 7} = 18$$

$$\Rightarrow x = \frac{35 \times 18}{2}$$

$$\Rightarrow x = 315$$

Now,

$$60\% \text{ of } 315 = 315 \times \frac{60}{100}$$

$$\Rightarrow 3 \times 63 = 189$$

$\therefore$ The answer is 189.

Question 41

The average weight of 50 boys in a class is 45 Kg. When one boy leaves the class, the average reduces by 100g. Find the weight of the boy who left the class.

Options:

- A. 50 Kg
- B. 50.8 Kg
- C. 49 Kg
- D. 49.9 Kg

Answer: D

Solution:

Given,

Average weight of 50 boys = 45 kg

Formula:

Average = sum of all observations / total number of all observations

Calculation:

Average weight of 50 boys = 45 kg

Sum of weight of 50 boys = $45 \times 50 = 2250$ kg

If 1 boy left the class, then average of 49 boys = 44.9 kg

Sum of weight of 49 boys = $44.9 \times 49 = 2200.1$

$\therefore$ Weight of boy who left the class = $2250 - 2200.1 = 49.9$ kg

Alternate Solution

New average = $45 - 0.1 = 44.9$ kg

Weight of boy left = New average + Number of total boys $\times$ Average reduced

= $44.9 + 50 \times 0.1$

= $44.9 + 5$

= 49.9 kg

Question 42

An Aeroplane started 30 minutes later than the scheduled time from a place 1500 km away from its destination. To reach the destination at the scheduled time the pilot needed to increase the speed by 250 km/hr. What was the speed of the Aeroplane during journey? (in km/hr).

Options:

A. 750

B. 1000

C. 800

D. 900

Answer: B

Solution:

Calculation:

Let the usual time taken by the Aeroplane = x hr.

Distance = 1500km

In first case-

speed = $1500/x$ km/h

In second case-

time taken by Aeroplane = $x - 1/2$ hr

speed = $\left(\frac{1500}{x - \frac{1}{2}}\right)$

Increasespeed = 250km/hr

$$\left(\frac{1500}{x - \frac{1}{2}}\right) - \frac{1500}{x} = 250$$

$$\Rightarrow \frac{1500}{1} \times \frac{2}{2x-1} - \frac{1500}{x} = 250$$

$$\Rightarrow \frac{3000x - 3000x + 1500}{(2x-1)x} = 250$$

$$\Rightarrow \frac{1500}{2x^2 - x} = 250$$

$$\Rightarrow 6 = 2x^2 - x$$

$$\Rightarrow 2x^2 - x - 6 = 0$$

$$\Rightarrow 2x^2 - 4x + 3x - 6 = 0$$

$$x = 2, x = -3/2$$

so, time can not be negative $x = 2$

$$\text{speed} = \frac{1500}{x - \frac{1}{2}} = \frac{1500}{2 - \frac{1}{2}} = 1000 \text{ km/h}$$

Question 43

**A number is first increased by 15% and it is further increased by 30%.
The original number is increased altogether by:**

Options:

A. 49%

B. 43%

C. 49.5%

D. 40%

Answer: C

Solution:

Given:

A number is first increased by 15% and it is further increased by 30%.

Concept used:

Effective increment = $a + b + \frac{a \times b}{100}$, where a and b are the percentage increase or decrease.

Calculation:

$$\text{Effective increment} = a + b + \frac{a \times b}{100}$$

According to the question,

$$a = 15\%, b = 30\%$$

$$\Rightarrow \text{total increment} = 15 + 30 + 15 \times 30/100$$

$$\Rightarrow \text{total increment} = 45 + 4.5 = 49.5\%$$

$\therefore$ altogether increment in number = 49.5% is Answer.

Question 44

How many members are nominated to the Rajya Sabha by the President of India?

Options:

A. 12

B. 14

C. 16

D. 10

Answer: A

Solution:

The correct answer is 12.

Important Points

- **12 members** could be nominated by the President of India in the Rajya Sabha. So, option 1 is correct.
- The **Rajya Sabha** should consist of not more than 250 members - 238 members representing the States and Union Territories, and 12 members nominated by the President.

Additional Information

- **Leader of the House:**
 - **The senior-most minister**, who is a member of Rajya Sabha, is appointed by the Prime Minister.
 - **Piyush Goyal** is the current Leader of the house.
 - **Deputy Chairman:**
 - Elected by the house from among its members.
 - The current Deputy Chairman of the Rajya Sabha is **Harivansh Narayan Singh**.
 - The current speaker of the Lok Sabha is **Om Birla** of the Bhartiya Janata Party.
 - The current chairman of the Rajya Sabha is **Jagdeep Dhankar**.
 - The Vice President of India is the ex-officio Chairman of Rajya Sabha.
-

Question 45

Which of the following is the hottest planet of our solar system?

Options:

A. Mercury

B. Mars

C. Venus

D. Jupiter

Answer: C

Solution:

The correct answer is Venus.

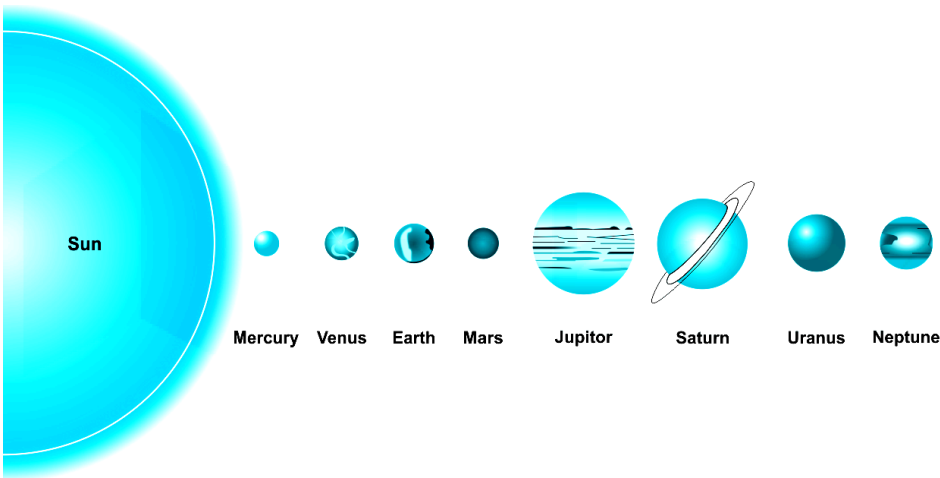
- Venus is the **hottest planet** in our solar system.
- This **planet inhibits** a very dense **atmosphere**.

Key Points

- Generally, the planets **which are nearest** to the **sun are hotter**.
- But the **planet Venus** is an exception to this.
- Venus has a **surface temperature** of about **888 degrees F**.
- Earth has a **surface temperature** of **61 degrees F**.
- **When the distance increases** from the sun, the **surface temperature** of the **planet decreases**.
- The **dense atmosphere** contributes to the heating of Venus by acting like a **greenhouse**.

Hint

- The **planet mercury** is **closest to the sun**.
- But **mercury is not the hottest planet**.



Question 46

In Women's Cricket, which country lifted the ICC T20 World Cup Trophy for the sixth time in February 2023?

Options:

- A. Pakistan
- B. Sri Lanka
- C. Australia
- D. Bangladesh

Answer: C

Solution:

The correct answer is Australia.

In News

- In Women's Cricket, **Australia** lifted the ICC T20 World Cup Trophy for the sixth time.

Key Points

- In the final match on 26 Feb 2023, they defeated **South Africa by 19 runs at Newlands, Cricket Ground in Cape Town**.
- Earlier, winning the toss and deciding to bat first, Australia posted 156 for 6 in their allotted 20 overs.
- Beth Mooney top-scored with unbeaten **74 runs for Australia**.

Additional Information

- **Australia:**
 - **Capital** - Canberra
 - **Currency** - Australian dollar
 - **Prime minister** - Anthony Albanese.
 - **National Sport** - Australian Rules football.
- **Other Sports News:**
 - England Test captain **Ben Stokes** became the player to hit the most number of sixes in the history of Test cricket in the match against New Zealand on 18 February 2023.
 - Jean-Eric Vergne of team **DS Penske** won the first-ever Formula E race held in Hyderabad on 11 February 2023.
 - With his 36th point against the **Oklahoma City Thunder** on February 7, 2023, LeBron James became the National Basketball Association's all-time leading scorer.
 - **Aditi Ashok** clinched the Ladies European Tour Title with a nine-shot win, despite 1-over on final day, at the Magical Kenya Ladies Open on 5 February 2023.
 - **Prajnesh Gunneswaran** won the final of the \$25,000 ITF men's tennis tournament in Al Zahra, Kuwait, on 29 January 2023.
 - England captain **Ben Stokes** has been named the International Cricket Council Test Player of the Year for 2022.

Question 47

Night blindness is caused due to the deficiency of

Options:

- A. vitamin A
- B. vitamin C
- C. vitamin B
- D. vitamin D

Answer: A

Solution:

The correct answer is **Vitamin A.**

Explanation:

- **Night blindness** or **Nyctalopia** is caused due to a deficiency of **Vitamin A**.
- Vitamin A is also called **Retinol**
- In this disease, the patient suffers from poor vision at night. Also, it may cause premature death in children below the age of 5.
- Food rich in vitamin a are Carrots, Spinach, Sweet potato, Lettuce, etc.



Additional Information

- **Vitamin D** is also known as **Calciferol**. It helps in the absorption of calcium dissolved in the blood, It prevents the weakening of bone and tooth.
 - Vitamin D also has other functions in the body, including cell growth, neuromuscular and immune function, and reduction of inflammation. Vitamin D deficiency can cause rickets in children and osteomalacia in adults.
 - **Vitamin B** is complex comprising of different vitamins: **B1 (thiamine)**, **B2 (riboflavin)**, **B3 (niacin)**, **B5 (pantothenic acid)**, **B6 (pyridoxin)**, **B7(biotin)**, **B9 (folate)** and **B12 (cobalamin)**.
 - **Vitamin C** is also known as **Ascorbic acid**. Its deficiency leads to the disease "Scurvy"
-

Question 48

In 2023 which of the following countries G7 summit was held?

Options:

- A. Japan
- B. Canada
- C. France
- D. Germany

Answer: A

Solution:

The Correct Answer is **Japan.**

In News

- **Japanese Prime Minister Fumio Kishida** welcomed the G7 leaders in **Hiroshima** where G7 leaders laid wreaths at the Peace Memorial Park.

Key Points

- The G7 leaders are currently in Japan to attend the **G7 Summit scheduled to be held in Hiroshima** from May 19-21.
- Japan assumed the G7 Presidency in 2023.
- **Hiroshima Peace Memorial Park** is the only structure left standing in the area, depicting the aftermath where the **world's first atomic bomb** was dropped on the city on **August 6, 1945**.
- Japan under its G7 Presidency has invited leaders from **Australia, Brazil, Comoros, Cook Islands, India, Indonesia, South Korea and Vietnam**.

Additional Information

- **Group of Seven (G7)**
 - It is an intergovernmental organization that was formed in **1975**.
 - The G7 countries are the UK, Canada, France, Germany, Italy, Japan and the US.
 - All the G7 countries and India are a part of the G20.
 - The G7 does not have a formal constitution or a fixed headquarters.
-

Question 49

What is called as the brain of computer and is responsible for processing data?

Options:

- A. Motherboard
- B. Memory
- C. Gateway
- D. CPU

Answer: D

Solution:

The correct answer is CPU.

Key Points

- **TheCPU**
 - **It is the brain of a computer, containing all the circuitry needed to process input, store data, and output results. Hence, Option 4 is correct.**
 - The CPU is constantly following instructions of computer programs that tell it which data to process and how to process it.
 - Without a CPU, we could not run programs on a computer.
 - The CPU can process those instructions easily, thanks to a control unit that knows how to interpret program instructions and an Arithmetic Logic Unit (ALU) that knows how to add numbers. With the control unit and ALU combined, the CPU can process much more complex programs than a simple calculator.
 - At the hardware level, a CPU is an **integrated circuit**, also known as a **chip**.
 - An integrated circuit "integrates" millions or billions of tiny electrical parts, arranging them into circuits and fitting them all into a compact box.

Additional Information

- **Motherboard**
 - The **motherboard** is a printed circuit board and foundation of a computer that is the biggest board in a computer chassis.
 - It allocates power and allows communication to and between the CPU, RAM, and all other computer hardware components.
- **Memory**
 - In computing, memory is a device or system that is used to store information for immediate use in a computer or related computer hardware and digital electronic devices.
 - The term memory is often synonymous with the term primary storage or main memory. An archaic synonym for memory is store.
- **A gateway**
 - It is a piece of networking hardware or software used in telecommunications for telecommunications networks that allows data to flow from one discrete network to another.

Question 50

Who among the following was the first to explain that the rotation of the earth on its own axis accounts for the daily rising and setting of the sun?

Options:

- A. Aryabhatta
- B. Bhaskara
- C. Brahmagupta
- D. Varahamihira

Answer: A

Solution:

The correct answer is Aryabhatta.

- Aryabhata was the first to explain that the rotation of the earth on its own axis accounts for the daily rising and setting of the sun.



Key Points

- Aryabhatta was born in India in 476 AD.
- He also stated that the Earth was a **spherical ball**.
- It takes around **23 hours, 56 minutes, and 4 seconds** to rotate on its own axis.



Additional Information

- India's first satellite was named after Aryabhatta.
 - It was launched with the help of USSR.
 - **Rohini** was the first satellite which was launched by India **onboard SLV-3**.
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